PRN:-202501120022

Practical1

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1.1.1. Calculate Momentum

18:57

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:
m is the mass of the object (in kilograms).
v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases +

calculate... Submit

```python
m=float(input(""))
v=float(input(""))
p=m*v
print(f"{p:.2f}kgm/s")
```

Terminal Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits

24:19

Write a Python program that accepts an integer n as input. Depending on the number of digits in n.

Constraints:
$1 \le n \le 999$

Input Format:

- The input consists of a single integer n.

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases +

condition... Submit

```python
import math
n=int(input(""))
if abs(n)<10:
	result = n**2
	print(result)
elif abs(n)<100:
	result = math.sqrt(n)
	print(f"{result:.2f}")
elif abs(n)<1000:
	result = n**(1/3)
	print(f"{result:.2f}")
else:
	print("Invalid")
```

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1.1.3. Days between Two Dates

22:29

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format $YYYY-MM-DD$.
- The second line contains the second date in the format $YYYY-MM-DD$.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases +

noOfDay... Submit

```python
from datetime import datetime

date1 = input().strip()
date2 = input().strip()

d1 = datetime.strptime(date1, "%Y-%m-%d")
d2 = datetime.strptime(date2, "%Y-%m-%d")

diff = (d2 - d1).days

print(diff)
```

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1.1.4. Reverse a Number 10:43

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases +

reverseN... Submit

```python
n=int(input())
reversed_n = int(str(n)[::-1])
print(reversed_n)
```

Terminal Test cases

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1.1.5. Multiplication Table 04:31

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases +

multiplica... Submit

```python
n = int(input(""))
for i in range(1, 11):
    result=n*i
    print(f"{n} x {i} = {result}")



```

Terminal Test cases

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1.2.1. Pass or Fail 44:29

Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n, the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases +

passorFa... Submit

```python
n = int(input(""))
marks=list(map(int, input().split()))

if len(marks) != n:
    print("Invalid Input")
else:

    if any (mark < 40 for mark in marks):
        print("Fail")
    else:
        aggregate = sum(marks) / n
        print(f"Aggregate Percentage: {aggregate:.2f}")
        if aggregate> 75:
            grade = "Distinction"
            print(f"Grade: {grade}")
        elif 60 <= aggregate < 75:
            grade = "First Division"
            print(f"Grade: {grade}")
        elif 50 <= aggregate < 60:
            grade = "Second Division"
            print(f"Grade: {grade}")
        elif 40 <= aggregate < 50:
            grade = 'Third Division'
            print(f"Grade: {grade}")
        else:
            print("Fail")
```

Terminal Test cases

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